

Lecture 3: Generalised Hydrodynamics for Integrable Quantum Field Theory

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You can find these lecture notes and some exercises at:
<https://olallacastroalvaredo.weebly.com/pentegost-school-25.html>



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B. Bertini, M. Collura, J. De Nardis, and M. Fagotti, Transport in out-of-equilibrium XXZ chains: exact profiles of charges and currents, 1605.09790 (**for integrable quantum spin chains**)



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- More recently (2025) the two papers were awarded the **Frontiers of Science Award** by the International Congress of Basic Science for representing “a major breakthrough in their field”.

GHD in a Nutshell

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- Spectral Functions: The state of the system is described by **quasi-particle distribution functions** $\rho_p(\theta, x, t)$. This is the same function as in TBA but is now **space-time dependent**.

- Continuity Equation: GHD describes evolution via a **continuity equation** : $\partial_t \rho_p(\theta, x, t) + \partial_x [v_{\text{eff}}(\theta, x, t) \rho_p(\theta, x, t)] = 0$ also

written as $\partial_t n(\theta, x, t) + v_{\text{eff}}(\theta, x, t) \partial_x n(\theta, x, t) = 0$ where v_{eff} is the **effective velocity of quasiparticles**, which is itself determined self-consistently via a TBA-like equation

$$v_{\text{eff}}(\theta, x, t) = v_{\text{gr}}(\theta) + \int d\alpha \frac{\varphi(\theta - \alpha) \rho_p(\alpha, x, t)}{p'(\theta)} (v_{\text{eff}}(\alpha, x, t) - v_{\text{eff}}(\theta, x, t))$$

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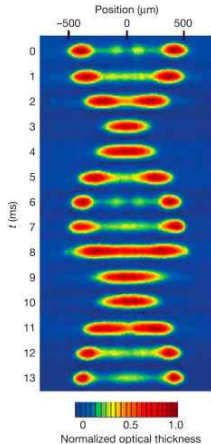
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- A particular example which has received a lot of attention is the problem of describing **equilibration after release from a trap**. This is the setting of a, by now, famous experiment: the **quantum Newton's cradle** experiment.

The Quantum Newton Cradle Experiment

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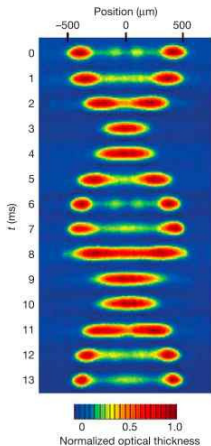
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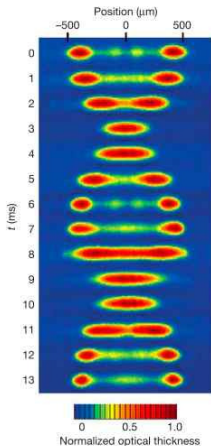
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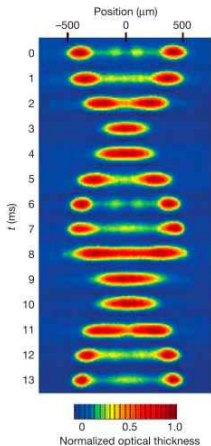
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- “Taking into account the trapping potential $V(x)$, the GHD equation reads

$$\partial_t \rho_p + \partial_x [v_{\text{eff}} \rho_p] = m^{-1} \partial_x V \partial_\theta \rho_p$$

Caux, Doyon, Dubail, Konik & Yoshimura'17

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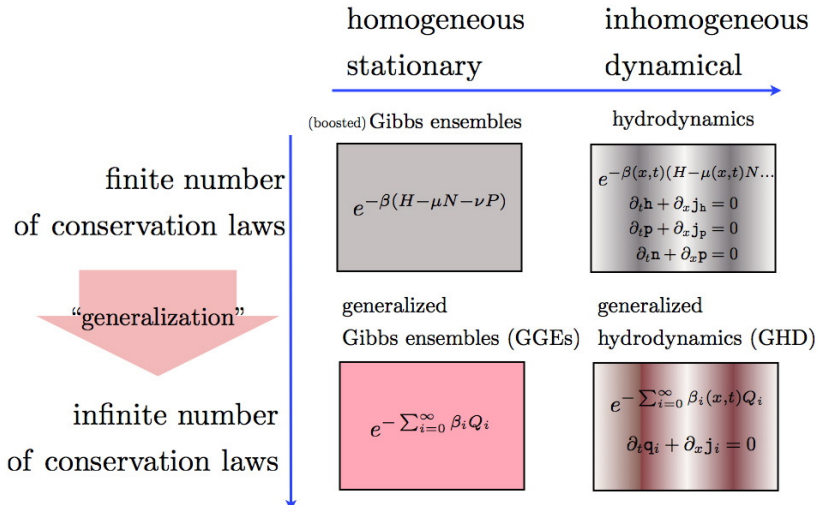
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- We now have **dynamics**. How?! Why?!

Generalised Hydrodynamics: Ambition



[© B. Doyon]

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$$\partial_t \mathbf{q}_s + \partial_x \mathbf{j}_s = 0, \quad \mathbf{q}_s, \mathbf{j}_s = \frac{\text{Tr} [\rho_{\text{GGE}}(x, t) \mathbf{q}_i, \mathbf{j}_s]}{\text{Tr} \rho_{\text{GGE}}(x, t)}, \quad s = 0, 1, 2, \dots$$

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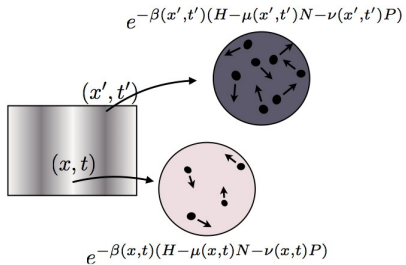
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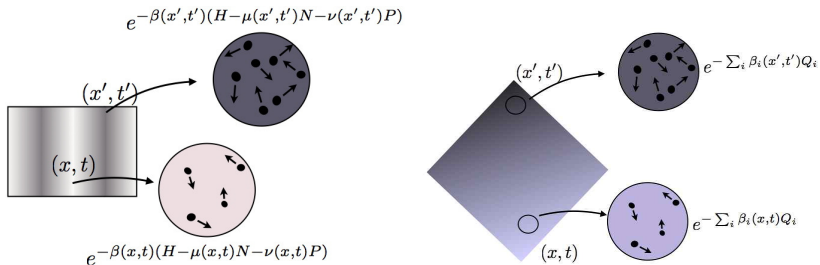
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- Conservation equations are **dynamical equations** determining the space-time dependent GGEs.

Microscopic/Mesososcopic/Macroscopic



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[© B. Doyon]

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$$v^{\text{eff}}(\theta; x, t) = \frac{(e')^{\text{dr}}(\theta; x, t)}{(p')^{\text{dr}}(\theta; x, t)}$$

- Then, the **conservation equations** can be rewritten as (the one-particle eigenvalues $h_s(\theta)$ are space-time independent).

$$\partial_t \rho_p(\theta; x, t) + \partial_x [v^{\text{eff}}(\theta; x, t) \rho_p(\theta; x, t)] = 0$$

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- In the context of GHD/TBA is that **the normal modes are the occupation functions** $n(\theta; x, t)$ and so the GHD equations in the quasi-particle picture can be written as

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- What this means is that $n(\theta; x, t) = n^0(u(\theta; x, t); x)$ for some function $u(\theta; x, t)$ that can be fixed by consistency. This is known as the **method of characteristics**.

Partitioning Protocol

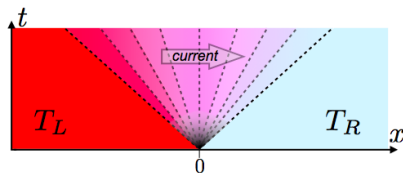
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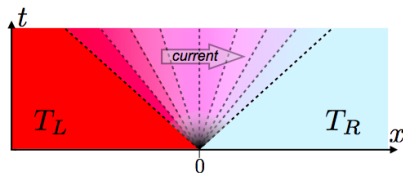
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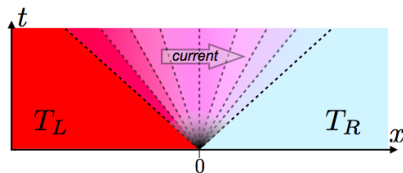
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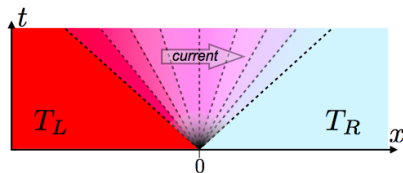
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Solution of Partitioning Protocol

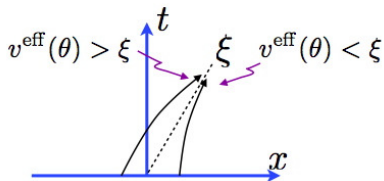
$$n(\theta; \xi) = n_L(\theta) \Theta(\theta - \theta_{\xi}) + n_R(\theta) \Theta(\theta_{\xi} - \theta), \quad v^{\text{eff}}(\theta_{\xi}; \xi) = \xi$$

Interpretation

- Particles originally on the left (hotter) reservoir, are originally thermalised at temperature T_L and propagate towards the right with their effective velocity and occupation function $n_L(\theta)$.

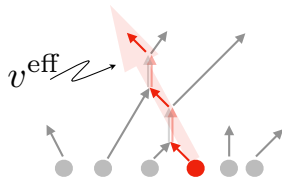
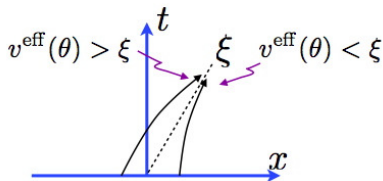
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- A TBA-like equation for $v^{\text{eff}}(\theta)$ can also be written, which is one of the important consequences of GHD:

TBA/GHD Equations for Effective Velocities

$$v^{\text{eff}}(\theta) = \frac{e'(\theta)}{p'(\theta)} + \frac{1}{p'(\theta)} \int_{-\infty}^{\infty} d\theta' \varphi(\theta - \theta') \rho_p(\theta') (v^{\text{eff}}(\theta') - v^{\text{eff}}(\theta))$$

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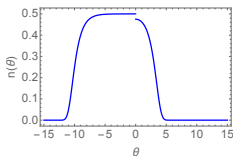
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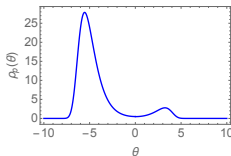
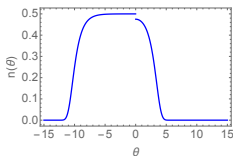
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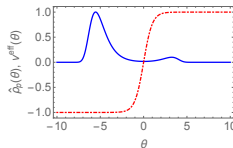
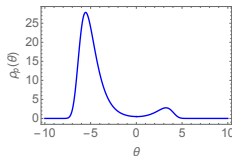
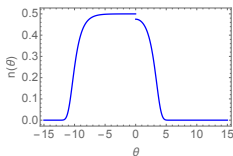
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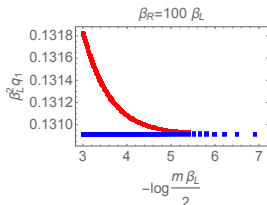
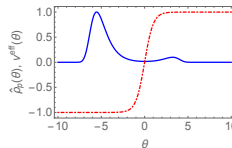
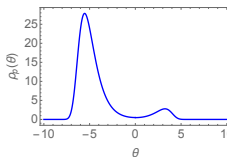
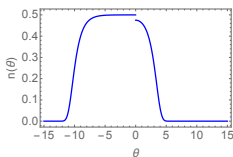
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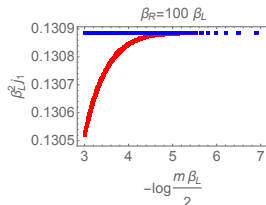
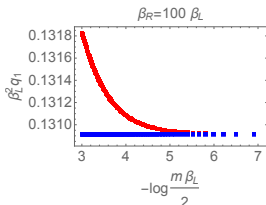
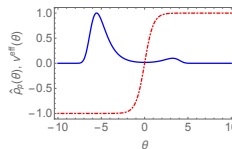
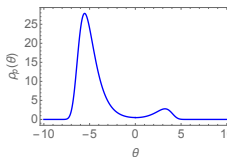
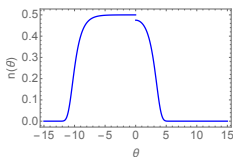
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Some References

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